

ERASTEEL TECHNICAL DATASHEET: BlueTap Max

Application Family: Cutting tools | Standards: Erasteel BlueTap Series PM HSS

1. Grade Overview & Characteristics

BlueTap Max is Erasteel's dedicated PM grade engineered specifically for threading and tap manufacturers. Delivers optimized grindability, chip clearance durability, and resilience against micro-chipping in internal thread tapping.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	0.90 - 0.96	4.0 - 4.4	4.8 - 5.2	6.1 - 6.5	1.7 - 1.9	4.6 - 5.0	Controlled micro-grain carbide structure

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	64.0 - 66.0 HRC	Machinability	7.0/10
Max Reachable Hardness	66.0 HRC	Wear Resistance	7.5/10
Annealed Delivery State	Max 280-350 HB	Toughness	7.0/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	7.5/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	8.0/10

4. Industrial Applications

High-performance machine taps, roll form taps, threading dies, shank tools for stainless steels and titanium threading.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1160-1200°C. Triple tempering at 560°C. Working hardness up to 66 HRC.